Linked Micromap Plots in R

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Abstract

Symanzik and Carr (2008, p. 284) indicated that no implementation of linked micromap (LM) plots in R existed at the time they wrote their book chapter in 2006. This considerably changed during the last seven years. First, multiple R scripts in support of Carr and Pickle (2010) have been developed and made available (`http://mason.gmu.edu/~dcarr/Micromaps/`). Recently, two full R packages, "micromap" (Payton and Olsen, 2013) and "micromapST' (Carr and Pearson Jr., 2013), have been developed. Moreover, additional micromap examples have been implemented in R by various authors. In this article, we will provide a comparative overview of these approaches and give some additional insights how to further modify existing R code to create individual new LM plots.

Keywords: geographic data, LM plots, software, statistical maps, visualization.

1. Introduction

Linked micromap (LM) plots have been introduced in the mid–1990s (Olsen et al., 1996; Carr and Pierson, 1996; Carr et al., 1998, 2000) as a new plotting method that combines row–labeled plots with a series of small maps. LM plots and their designs evolved over time. Comprehensive overviews and examples of LM plots can be found in Symanzik and Carr (2008) and Carr and Pickle (2010). LM plots are frequently used to display relatively simple statistical summaries in a geographic setting, such as states, counties, ecoregions, and others. LM plots can be understood relatively easily by the general public without any advanced statistical background knowledge and, therefore, they are a valuable statistical tool to communicate spatial information to varied audiences.

Originally, LM plots were primarily created in S–Plus and in Java. In their book chapter on LM plots, written in 2006, Symanzik and Carr (2008, p. 284) indicated: "It seems to be possible to adapt the S–Plus micromap code for use in the R statistical package (Ihaka and Gentleman, 1996), which can be freely obtained from `http://www.r-project.org/`. Although no full implementation of a micromap library in R exists at this point, the basic panel functions and simple examples of LM plots have been converted from S–Plus to R by Anthony R. Olsen." Since the writing of this book chapter, things changed considerably during the last seven years. A series of R (R Core Team, 2012) scripts has been created in support of Carr and Pickle (2010), two full R packages, "micromap" (Payton and Olsen, 2013) and "micromapST' (Carr and Pearson Jr., 2013), have been developed, and several other authors have developed stand–alone solutions for LM plots. In the next four sections of this article, we will introduce various approaches that allow the creation of LM plots in R. We finish with a brief discussion and outlook in Section 6.

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2. LM Plot Templates in Support of Carr and Pickle (2010)

The web site at `http://mason.gmu.edu/~dcarr/Micromaps/` supports the book by Carr and Pickle (2010). The site includes R functions and scripts for linked micromap plots and comparative micromap plots and these can be freely accessed. Numerous sample data sets and boundary files allow the recreation of LM plots from the book. The documents provide background information how to prepare additional data and boundary files for different geographical, political, economical, or environmental regions.

Examples cover LM plots at the states level of the United States (US), counties or parishes of selected US states, and specific subregions such as the north–central US or the southern US, or even locations on a world map. Multiple plot types for the data panels are supported, such as dot plots, means and confidence intervals, bar charts, time series plots, and immigration plots.

A central part of the implementation is a set of panel functions, such as panelLayout, panelSelect, panelScale, panelFill, and panelGrid. These functions allow users to specify details of the overall plotting area of each component of the plot — and then draw the desired subplot into this plotting area. Technically, this means that a user has to loop through each of the various panels (for maps, identifiers, and statistical data) and each of the rows of the data and explicitly indicate which information (dots, names, regions in a map) has to be plotted in that panel.

Unfortunately, these scripts require some advanced R knowledge, even if a user only wants to make small modifications in the provided code. Otherwise, a user has full control, allowing changes of even the smallest details in the appearance of the resulting LM plots.

3. The R Package "micromap"

Others noticed similar difficulties with the LM plot templates from Carr and Pickle (2010). Payton et al. (2012) stated: "Producing LMplots plots has typically been somewhat difficult, and therefore LMplots have seen limited use." Therefore, one of the objectives of this group of researchers was: "Create an R library that handles the layout and arrangement of LMplots given simple input of GIS data and statistical summaries." The result is the R package "micromap" (Payton and Olsen, 2013). An example LM plot, based on this package, is shown in Figure 1.

Rather than using panel functions as the R scripts in support of Carr and Pickle (2010), the "micromap" package is based on the R package "ggplot2" (Wickham, 2009; Wickham and Chang, 2013). This has the advantage that existing R functionaly from "ggplot2" can be reused. From a user's perspective, it is not necessary to loop through each column and row of the data to create a cell in the LM plot. Rather, this functionality is hidden in two of the main functions of this R package: "lmplot" and "lmgroupedplot". A user basically has to provide the data for the plot and maps and specify the underlying layout of the LM plot as arguments of these two functions. Changes in the design can be made relatively easily, without any in–depth R programming knowledge. Payton et al. (2013b) state: "5 and 7 point summary box plots, dot plots, and bar plots are included by default. However, the package is designed to allow a user to create their own desired statistical graph to be included with minimal effort." Further details on the functionality of this R package and numerous examples can be found in Payton et al. (2012), Payton et al. (2013b), and Payton et al. (2013a).

4. The R Package "micromapST"

Pickle et al. (2013) describe the R package "micromapST" (Carr and Pearson Jr., 2013) that has been recently submitted to CRAN. This package focuses exclusively on special linked map design for U.S. states. It extends and replaces the local package "stateMicromap" used in statistical graphics classes at George Mason University (GMU) since the Fall of 2011 because of its simplified input. The package is still essentially based the same panel layout functions described in Carr (1997) and converted to R by Anthony R. Olsen. The input simplification replaces writing or editing a long linked micromap script, such as the dot plot confidence interval example (Carr, 1997). The two required input arguments are a data frame with a row for each state

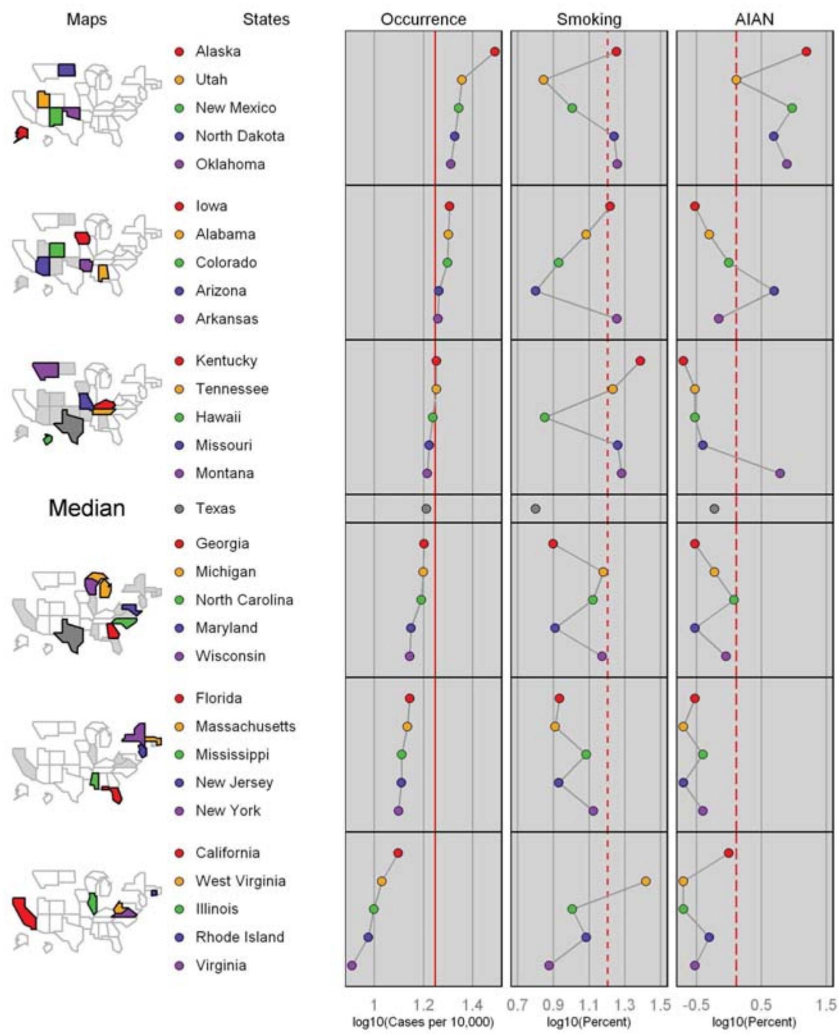

Figure 1: Close reproduction of Figure 2 from Gebreab et al. (2008), based on the "micromap" R package.

and a panel description data frame with a row for each column in the linked micromap. An example LM plot, based on this package, is shown in Figure 2.

The focus in constructing LM plots via this package is on constructing the panel description data frame. The columns indicate the type of map or panel glyph to use, the plot labeling, and the columns to use from the state data frame. The "dotconf" glyph needs to have the columns for the state estimates, and lower and upper bounds. Some glyphs, such as box plots, time series, scatterplots, and QQ–plots, require input data structures such as lists and arrays. An additional column called "panelData" provides for specifying the data structure names. Further details on the functionality of this R package and numerous examples can be found in Pickle et al. (2013).

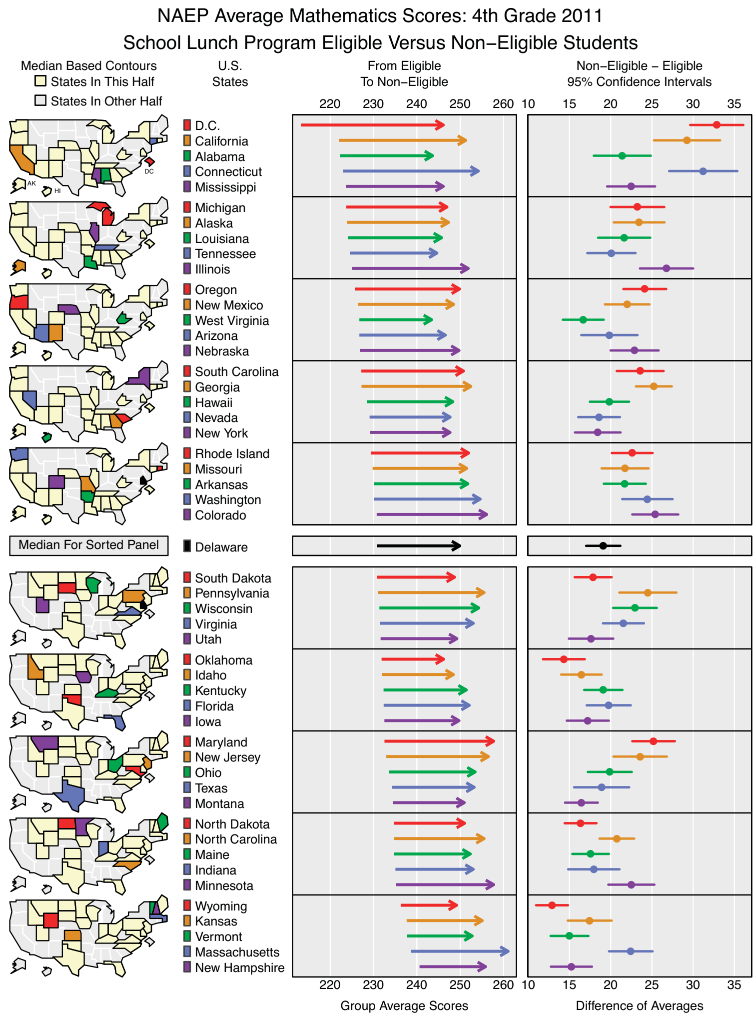

Figure 2: LM plot, based on the “micromapST” R package.

5. Stand–Alone LM Plot Approaches

A few additional attempts have been made to create LM plots in R. Perhaps one of the earliest implementations of a working version of LM plots in R is due to Mike Minnotte. He created a micromap script for his graduate–level “Graphical Methods” course at Utah State University (USU) around 2006/2007. His implementation is based on the R command “layout” and the R package “maps” (Becker et al., 2013). His code is still being used for teaching purposes at USU today.

In 2011, Brian Diggs posted some outline how to create LM plots via the R packages “ggplot2” and “maps.” His code, accessible at `https://groups.google.com/forum/?fromgroups#!topic/`

ggplot2/djLY7AeCd7Y, only has limited functionality, but it outlines the overall mechanism behind LM plots quite well.

For his MS project at USU in 2011/2012, Nathan Voge worked with the micromap R code from Mike Minnotte and the R code created in support of Carr and Pickle (2010). In particular, he further documented the original code to make it easier for others to better understand the original code — and eventually make adaptations to this code for new data sets. His resulting LM plots (and the commented code) can be found in Voge and Symanzik (2011) and Voge (2012).

6. Conclusion

Certainly, the easiest way to create LM plots in R is via the R packages "micromap" and "micromapST." Many historical micromaps that were originally created in S–Plus and in Java can be easily reproduced via these R packages. However, the R functions in both packages only have a limited number of arguments. If a user needs to create a new LM plot design not readily supported by the "micromap" and "micromapST" packages, it may be beneficial to work directly with te R code created in support of Carr and Pickle (2010).

In the last few years, regional linked micromap plots for France (Bonnal et al., 2011), and, more recently, for Korea (Ahn, 2013), have been created. It can be expected that the R packages "micromap" and "micromapST" will help to encourage and facilitate the production of regional linked micromap plots and introduce such plots to many new scientists and practitioners.

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